Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 30 - Columns: 25 - Key Features:
- Customer demographics (Age, Gender, Location, Membership Status)
- Purchase details (Item Purchased, Category, Items Purchased, Average Spend, Payment Method)
- Shopping behavior (Online Shopping, Discount Used, Previous Purchases, Repeat Buyers, Shipping Type, Purchase Status, Coupon Code, Review Rating, Delivery Days, Satisfaction Rating)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer_ID	Age	Gender	City	Membership	Visit_Frequency	Preferred_Category	Average_Spend	Payment_Method	Online_Shopping
count	30	30.000000	30	30	25	30	30	30.000000	30	30
unique	30	NaN	2	12	3	3	8	NaN	4	2
top	C001	NaN	Female	Manila	Gold	Weekly	Electronics	NaN	Credit Card	Yes
freq	1	NaN	15	3	10	14	5	NaN	9	21
mean	NaN	33.100000	NaN	NaN	NaN	NaN	NaN	4566.666667	NaN	NaN
std	NaN	7.707363	NaN	NaN	NaN	NaN	NaN	3127.005104	NaN	NaN
min	NaN	21.000000	NaN	NaN	NaN	NaN	NaN	1200.000000	NaN	NaN
25%	NaN	27.000000	NaN	NaN	NaN	NaN	NaN	2325.000000	NaN	NaN
50%	NaN	32.500000	NaN	NaN	NaN	NaN	NaN	3400.000000	NaN	NaN
75%	NaN	38.750000	NaN	NaN	NaN	NaN	NaN	5975.000000	NaN	NaN
max	NaN	47.000000	NaN	NaN	NaN	NaN	NaN	12500.000000	NaN	NaN

Discount_Used	Items_Purchased	Satisfaction_Rating
30	30.000000	30.000000
2	NaN	NaN
Yes	NaN	NaN
18	NaN	NaN
NaN	4.966667	4.533333
NaN	3.872835	0.507416
NaN	1.000000	4.000000
NaN	2.250000	4.000000
NaN	4.000000	5.000000
NaN	5.000000	5.000000
NaN	15.000000	5.000000

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:** Verified if discount_used and coupon_code were redundant; dropped coupon_code.

- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	90
2	Male	59

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id text	items_purchased bigint
1	C001	5
2	C007	6
3	C009	10
4	C010	5
5	C017	5
6	C021	5
7	C023	6
8	C027	5

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type character varying (30)	round numeric
1	Standard	4.43
2	Express	4.67

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	membership text	total_customers bigint	average_spend numeric	total_revenue numeric
1	Gold	10	6660.00	66600.00
2	Silver	9	5066.67	45600.00
3	Bronze	6	2533.33	15200.00
4	[null]	5	1920.00	9600.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	items_purchased bigint	discount_rate numeric
1	10	100.00
2	6	100.00
3	5	100.00
4	4	100.00
5	3	66.67

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Returning	14
2	Loyal	13
3	New	3

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category character varying (100)	item_purchased character varying (100)	total_orders bigint
1	1	Beauty	Headphones	2
2	2	Beauty	Jacket	1
3	1	Books	Headphones	1
4	2	Books	Football	1
5	1	Electronics	Jacket	2
6	2	Electronics	Shoes	2
7	3	Electronics	Keyboard	1
8	4	Electronics	Laptop	1
9	5	Electronics	Headphones	1
10	1	Groceries	Shoes	1
11	2	Groceries	Basketball	1
12	3	Groceries	Laptop	1
13	1	Home	Football	1
14	2	Home	Keyboard	1
15	1	Sports	Laptop	2
16	2	Sports	Basketball	1
17	3	Sports	Keyboard	1
18	4	Sports	Mouse	1
19	5	Sports	Watch	1
20	6	Sports	Backpack	1
21	7	Sports	Shoes	1
22	1	Toys	Mouse	2
23	2	Toys	Keyboard	1
24	3	Toys	Backpack	1
25	4	Toys	Shoes	1

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	membership text	repeat_buyers bigint
1	[null]	4
2	Bronze	4
3	Silver	8
4	Gold	5

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_category text	total_revenue numeric
1	Adult	79000
2	Middle-aged	45200
3	Young Adult	12800

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

6. Business Recommendations

- **Boost Membership** – Promote exclusive benefits for memberships.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the highest segment and depends on what membership avails.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.